A/B Study — Participant Data Collection Sheet

About this study

Thank you for taking part in this research. We are studying whether short explanations can help developers diagnose why an AI safety guardrail made a wrong decision — either flagging a safe prompt as unsafe, or missing a harmful one.

You will review 25 guardrail failure cases per session across two sessions (at least 3 days apart). For each case, identify the most likely root cause and suggest a fix. There are no trick questions — we want your honest judgment.

Each case takes about 1–2 minutes. Use a phone stopwatch to record your time. Total session time: ~30–45 minutes.

Each case shows you a guardrail decision that went wrong. Some cases include additional analysis, some don't — just work with whatever information is on the page.

Content notice: Cases are from the ToxicChat dataset (huggingface.co/datasets/lmsys/toxic-chat). Some prompts contain foul or offensive language — this reflects real-world safety content. You may stop and withdraw at any time if you feel uncomfortable.

Participant consent

This study collects only your answers and the time you spend on each case. No personally identifiable information is stored — your responses are recorded under a participant ID only (e.g. P01). Your data will be used solely for academic research and will not be shared with any third party. You may withdraw at any time without penalty.

☐ I agree to take part. I understand only anonymised data is collected and I can withdraw at any time.

Signature: ______________________ Date: ______________

Participant ID	Role	LLM familiarity (1–5)	Date

Q1 — Root Cause:
a) Over-triggered on surface wording b) Novel evasion / training gap d) Misinformation risk e) Guard is correct (unsafe/safe) FP = false positive FN = false negative Co

Column legend — what to write in each column

Case	Case number from your booklet (1–25)
T	Type: write FP (false positive) or FN (false negative) — shown on each case page
Secs	How many seconds this case took (stopwatch lap time)
Q1	Root cause answer: write a, b, c, d, or e
Conf	Your confidence: 1=not confident 2=somewhat 3=confident 4=very confident
✓?	Leave blank — researcher fills this in after scoring

SESSION 1 — 25 cases

Form received

Session start

Session end

Case	T	Secs	Q1	Conf	✓?
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					
Totals →					

T: FP=false positive / FN=false negative | ✓? filled by researcher: 1=correct 0=incorrect

SESSION 2 — 25 cases (at least a couple of hours after Session 1)

Form received | Session start | Session end

Case	T	Secs	Q1	Conf	✓?
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					
Totals →					

T: FP=false positive / FN=false negative | ✓? filled by researcher: 1=correct 0=incorrect

SUMMARY (filled by researcher after scoring)

Metric	Session 1	Session 2
Arm (control / treatment)		
Mean diagnostic time (s)		
Total correct (/ 25)		
Accuracy (%)		
Mean confidence (1–4)		

Primary metric: mean diagnostic time (H1: ≥30% reduction, $p<0.05$) | Secondary: accuracy ≥85% in treatment arm